Counterexample to the Dual Span Question for Combinatorial Banach Spaces

FA/Banach run `fa_banach_001`

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Status

This packet gives a counterexample to Question 6.1 of Borodulin-Nadzieja, Farkas, Jachimek, and Pelczar-Barwacz, arXiv:2404.01733. The question asks whether, for every hereditary finite cover $\mathcal{F}$,

$$\overline{\mathrm{span}}(W(\overline{\mathcal{F}})) = X_{\mathcal{F}}^{*} \quad \text{or even} \quad \overline{\mathrm{conv}}(W(\overline{\mathcal{F}})) = B(X_{\mathcal{F}}^{*})$$

in norm.

Source Question

The source defines

$$W(\mathcal{H}) = \{\sigma \in \{\pm 1, 0\}^{\mathbb{N}} : \mathrm{supp}(\sigma) \in \mathcal{H}\} \subseteq B(X_{\mathcal{F}}^{*})$$

for $\mathcal{H} \subseteq \overline{\mathcal{F}}$ and records that $W(\mathcal{H})$ is weak-star norming when $\mathcal{F} \subseteq \mathcal{H}$.

If $f \in X_{\mathcal{F}}^{*}$ and $\alpha = (f(e_n)) \in \mathbb{R}^{\mathbb{N}}$, then we will write $f(x) = \langle \alpha, x \rangle$. It follows that if $\alpha, \beta \in \mathbb{R}^{\mathbb{N}}$, $\alpha \in X_{\mathcal{F}}^{*}$, and $|\beta(n)| \leq |\alpha(n)|$ for every n, then $\beta \in X_{\mathcal{F}}^{*}$ as well and $\|\beta\|_{\mathcal{F}}^{*} \leq \|\alpha\|_{\mathcal{F}}^{*}$ where $\| \bullet \|_{\mathcal{F}}^{*}$ stands for the usual norm of $X_{\mathcal{F}}^{*}$. (As a side remark, let us mention that the norm $\| \bullet \|_{\mathcal{F}}^{*}$ is a Banach envelope of a certain simply definable quasi-norm of combinatorial nature, see [10].)

If $\beta \in B(X_{\mathcal{F}}^{*})$ then $|\beta(n)| \leq 1$ for every n; conversely, if $\mathrm{supp}(\beta) \in \overline{\mathcal{F}}$ and $|\beta(n)| \leq 1$ for every n, then $\beta \in B(X_{\mathcal{F}}^{*})$. In particular, if $\sigma \in \{\pm 1, 0\}^{\mathbb{N}}$ is such that $\mathrm{supp}(\sigma) \in \overline{\mathcal{F}}$, then $\sigma \in B(X_{\mathcal{F}}^{*})$ and, unless $\mathrm{supp}(\sigma) = \emptyset$, $\|\sigma\|_{\mathcal{F}}^{*} = 1$. Now, if $\mathcal{H} \subseteq \overline{\mathcal{F}}$ define

$$W(\mathcal{H}) = \left\{\sigma \in \{\pm 1, 0\}^{\mathbb{N}} : \mathrm{supp}(\sigma) \in \mathcal{H}\right\} \subseteq B(X_{\mathcal{F}}^{*}).$$

Notice that if $\mathcal{F} \subseteq \mathcal{H}$ then $W(\mathcal{H})$ is a *norming set*, that is,

$$\|x\|_{\mathcal{F}} = \sup \left\{ |\langle \sigma, x \rangle| : \sigma \in W(\mathcal{H}) \right\} \text{ for every } x \in X_{\mathcal{F}},$$

equivalently (as $W(\mathcal{H}) = -W(\mathcal{H})$, see [11, Lemma 4]), $\overline{\mathrm{conv}}^{w^*}(W(\mathcal{H})) = B(X_{\mathcal{F}}^{*})$.

Also, one can easily check that the weak* topology on $W(\mathcal{H})$ coincides with the inherited topology from the product $\{\pm 1, 0\}^{\mathbb{N}}$; and it follows that $(W(\mathcal{H}), w^*)$ is compact iff $W(\mathcal{H}) \subseteq \{\pm 1, 0\}^{\mathbb{N}}$ is compact iff $\mathcal{H} \subseteq \mathcal{P}(\mathbb{N})$ is compact. It also follows

Question 6.1 asks whether the corresponding norm-closed linear or convex hulls already give the whole dual or its unit ball.

6. FURTHER QUESTIONS

Consider $\mathcal{F} \in$ FHC. Obviously, $\operatorname{conv}(W(\max(\overline{\mathcal{F}}))) \supseteq W(\overline{\mathcal{F}})$, therefore, regarding convex and linear hulls of $\operatorname{Ext}(B(X^*_{\mathcal{F}})) = W(\max(\overline{\mathcal{F}}))$, we can work with $W(\overline{\mathcal{F}})$ instead. The first question is motivated by the fact (see [3, Corollary 4.6]) that, assuming $\mathcal{F}$ is compact, $X^*_{\mathcal{F}}$ satisfies the λ-property (i.e. the CSRP).

Question 6.1. Does $\overline{\operatorname{span}}(W(\overline{\mathcal{F}})) = X^*_{\mathcal{F}}$, or even $\overline{\operatorname{conv}}(W(\overline{\mathcal{F}})) = B(X^*_{\mathcal{F}})$, always hold? If so, then we may go further: Does $X^*_{\mathcal{F}}$ always satisfy the λ-property?

In general, c_0-saturation of a Banach space does not imply ℓ_1-saturation of its dual (see [21]). However, the question is still open for the case of combinatorial Banach spaces, that is, for combinatorial spaces generated by compact families. Note

Counterexample

Let Ω be the disjoint union of finite blocks I_k with $m_k = |I_k| \to \infty$. Define

$$\mathcal{F} = \{E \in [\Omega]^{<\infty} : |E \cap I_k| \le 1 \text{ for every } k\}.$$

After identifying Ω with $\mathbb{N}$, this is an element of FHC: it is hereditary and covers Ω.

Theorem. *For this $\mathcal{F}$,*

$$\overline{\operatorname{span}}(W(\overline{\mathcal{F}})) \neq X^*_{\mathcal{F}}.$$

Consequently,

$$\overline{\operatorname{conv}}(W(\overline{\mathcal{F}})) \neq B(X^*_{\mathcal{F}}).$$

Proof Intuition

The family $\mathcal{F}$ allows only one point per block. Thus a selector-sign functional in $W(\overline{\mathcal{F}})$ has at most one nonzero coordinate in each block. A finite linear combination of r such functionals has at most r nonzero coordinates in each block. The dual unit ball nevertheless contains vectors that spread mass uniformly over every block. Since the block sizes tend to infinity, no uniformly bounded number of atoms can approximate these uniform block measures in ℓ_1 on every block at once.

Proof

For $x \in c_{00}(\Omega)$,

$$\|x\|_{\mathcal{F}} = \sup_{E\in\mathcal{F}} \sum_{\omega\in E} |x(\omega)| = \sum_{k=1}^{\infty} \|x|_{I_k}\|_\infty.$$

Indeed, a finite selector may choose one coordinate of maximum absolute value from each nonzero block of x, and no selector can do better inside a block. Therefore

$$X_{\mathcal{F}} = \left(\bigoplus_{k=1}^{\infty} \ell_\infty^{m_k}\right)_{\ell_1}.$$

Since the summands are finite-dimensional,

$$X_{\mathcal{F}}^* = \left(\bigoplus_{k=1}^{\infty} \ell_1^{m_k}\right)_{\ell_\infty}, \qquad \|\alpha\| = \sup_k \|\alpha|_{I_k}\|_1.$$

The closure $\overline{\mathcal{F}}$ consists exactly of all selectors $A \subseteq \Omega$ with $|A \cap I_k| \leq 1$ for every k, finite or infinite. Hence every $\sigma \in W(\overline{\mathcal{F}})$ has at most one nonzero coordinate in each block.

Define $u \in X_{\mathcal{F}}^*$ by

$$u(\omega) = \frac{1}{m_k} \quad (\omega \in I_k).$$

Then

$$\|u\| = \sup_k \sum_{\omega \in I_k} \frac{1}{m_k} = 1,$$

so $u \in B(X_{\mathcal{F}}^*)$.

Let $y = \sum_{j=1}^r a_j \sigma_j$ be any finite linear combination of elements of $W(\overline{\mathcal{F}})$. For every block I_k, the support of $y|_{I_k}$ has cardinality at most r, because each σ_j contributes to at most one coordinate in that block. Choose k with $m_k > 2r$. Then

$$\|u|_{I_k} - y|_{I_k}\|_1 \geq \sum_{\omega \in I_k \setminus \operatorname{supp}(y|_{I_k})} \frac{1}{m_k} \geq \frac{m_k - r}{m_k} > \frac{1}{2}.$$

Thus

$$\|u - y\|_{X_{\mathcal{F}}^*} = \sup_k \|u|_{I_k} - y|_{I_k}\|_1 > \frac{1}{2}.$$

This holds for every algebraic finite linear combination y, so u is not in the norm closure of $\operatorname{span}(W(\overline{\mathcal{F}}))$.

Since

$$\operatorname{conv}(W(\overline{\mathcal{F}})) \subseteq \operatorname{span}(W(\overline{\mathcal{F}})),$$

the same u is not in the norm-closed convex hull. This proves both negative assertions.

Verification Notes

The argument uses only the definitions from the source paper and standard duality for ℓ_1 sums. The obstruction is uniform in the number of terms of a proposed approximation: each finite combination fixes a finite support bound per block, and a larger block defeats that approximation by at least 1/2 in the dual norm.

Novelty Check

The local registry, solution index, attempt index, and proof-gap index were searched for the source arXiv id, Question 6.1, dual span, closed convex hull, the lambda-property, and related combinatorial-Banach terms. Web searches for the source title, the arXiv id, $W(\overline{\mathcal{F}})$, and the dual-ball span phrasing found the source paper but no later direct answer to Question 6.1.

Limitations

This counterexample refutes the general assertions in Question 6.1. It does not address whether compact hereditary families, or other restricted subclasses of combinatorial families, satisfy the norm-closed span or convex-hull property. It also does not independently settle the lambda-property for arbitrary $X_{\mathcal{F}}^*$.

Reference

P. Borodulin-Nadzieja, B. Farkas, S. Jachimek, A. Pelczar-Barwacz, *The Zoo of combinatorial Banach spaces*, arXiv:2404.01733.

Human Review: Selector-Block Dual-Span Counterexample

Original automatic proof: `solution_packet.pdf`
Checked date: 19 June 2026
Status: Manually checked.
Verdict: The proof appears correct.

Human Comments

The packet gives a counterexample to Question 6.1 of Borodulin-Nadzieja, Farkas, Jachimek, and Pelczar-Barwacz, concerning whether the norm-closed span, or even the norm-closed convex hull, of the sign functionals

$$W(\overline{\mathcal{F}})$$

recovers the full dual space or the full dual unit ball of the corresponding combinatorial Banach space.

The construction is clear and appears sound. One decomposes the underlying set into finite blocks

$$\Omega = \bigsqcup_{k=1}^{\infty} I_k, \qquad |I_k| = m_k \to \infty,$$

and takes $\mathcal{F}$ to be the hereditary family of finite selectors, namely finite sets meeting each block in at most one point. Its closure consists exactly of the possibly infinite selectors

$$A \subseteq \Omega, \qquad |A \cap I_k| \le 1 \quad (k \in \mathbb{N}),$$

so every element of $W(\overline{\mathcal{F}})$ is a signed selector with at most one nonzero coordinate in each block.

The identification of the associated space is correct:

$$\|x\|_{\mathcal{F}} = \sum_k \|x|_{I_k}\|_\infty,$$

and hence

$$X_{\mathcal{F}} = \left(\bigoplus_k \ell_\infty^{m_k}\right)_{\ell_1}, \qquad X_{\mathcal{F}}^* = \left(\bigoplus_k \ell_1^{m_k}\right)_{\ell_\infty}.$$

The dual norm is therefore

$$\|\alpha\| = \sup_k \|\alpha|_{I_k}\|_1.$$

The obstruction is the blockwise uniform vector $u \in X_{\mathcal{F}}^*$, defined by

$$u(\omega) = \frac{1}{m_k} \qquad (\omega \in I_k).$$

For every block, $\|u|_{I_k}\|_1 = 1$, so u belongs to the dual unit ball.

On the other hand, if

$$y = \sum_{j=1}^{r} a_j \sigma_j, \qquad \sigma_j \in W(\overline{\mathcal{F}}),$$

then in each block I_k, the vector $y|_{I_k}$ is supported on at most r coordinates, because each signed selector contributes to at most one coordinate in that block. Choosing k with $m_k > 2r$, one obtains

$$\|u|_{I_k} - y|_{I_k}\|_1 \geq \frac{m_k - r}{m_k} > \frac{1}{2}.$$

Therefore

$$\|u - y\|_{X_{\mathcal{F}}^*} > \frac{1}{2}$$

for every finite linear combination y of elements of $W(\overline{\mathcal{F}})$. This proves that u is not in the norm-closed linear span of $W(\overline{\mathcal{F}})$. Since the convex hull is contained in the linear span, the same vector also witnesses failure of the asserted norm-closed convex-hull equality.

The proof is short, but the mechanism is convincing: finite combinations of selectors are uniformly sparse in each block, whereas the dual unit ball contains vectors whose mass is spread over arbitrarily large blocks. The growth $m_k \to \infty$ is exactly what makes the obstruction uniform over all finite combinations.

Review Outcome

I would classify this as a correct counterexample to the general form of Question 6.1. It should be noted that the construction does not address possible positive results for more restrictive subclasses of hereditary families, such as compact families or other special combinatorial families. For the unrestricted formulation considered in the packet, however, the proof appears correct.